Division properties of exponents and [redacted] zero exponent

1

a 10^2 b j^4 c a^2 d $\frac{9}{4}q^4$

2

a 18^9 b a^4 c u^{11} d t^2r^4

e $5r^3$ f $\frac{3j^5}{4}$ g $\frac{-x^7}{3}$ h $\frac{9}{4}j^2k^8$

i $\frac{3g^3}{2h^6}$ j $9j^2k^3$ k $\frac{jk}{7}$ l $\frac{3}{8}mn^2$

m $2pq^2$ n $\frac{p^6q^3}{3}$ o $\frac{3j^2k^2}{2}$ p $\frac{y^4}{16}$

q $3rs^2$ r $-3x^9$ s $\frac{x^2}{4}$ t $9x^5$

3

a 2 b $3j^7$ c x^8y^5 d $g^{20}h^8$

e $7x^4$ f x^7y^5 g $4r^2s^6$

4

a $n^{20}r^{20}$ b $10x^2y^{11}z^{15}$ c $-5x^2y^{14}z^{11}$ d m^{100}

e $-6u^{11}$ f y^7 g $2p^{14}$ h $4x^2$

i $9v^5y^{11}$ j m^8 k p^5 l x^3

m $\frac{3b^5}{a}$ n $6p^7$

5 1

6

a p^0 b $\frac{b^{11}}{b^{11}} = b^0$ c $\frac{w^7}{w^7} = w^0$

7

a 1 b 1 c 1

8

a 18 b 1 c 11 d 1

e 1 f 9 g $27m^{13}$ h $5y^5$

Additional questions

9 No. The expression in the numerator disallows the fraction to be separated into two parts with a denominator of a^2. So the quotient rule cannot be applied to both parts of the numerator.

10 Yes. The expression in the numerator allows the fraction to be separated into two parts, each with a denominator of a^2. So the quotient rule can be applied to both terms.